



As the industry moves to 16-nanometre processes, HSPICE is ready with the industry's fastest and most accurate FinFET support, offering BSIM-CMG models with simulation speed comparable to that of planar CMOS — Hany Elhak, senior product marketing manager, Synopsys

misation, encryption, model editor, support for parametrised models, auto-convergence and checkpoint restart, several internal solvers, magnetic part editor, and Tabrizi core model support for non-linear cores.

PSpice simulates complex mixed-signal designs containing both analogue and digital parts. It supports a wide range of simulation models such as IGBTs, pulse-width modulators, digital-to-analogue converters and analogue-to-digital converters.

"Its built-in mathematical functions and behavioural modelling techniques enable fast and accurate simulation

that can do circuit DC, AC, transient, Fourier and noise analysis. TINA-TI is a simple version of TINA with full circuit simulation functionality but less circuit development utilities in the package. It also contains more TI device models and TI part information.

TINA-TI is ideal for designing, testing and troubleshooting a wide variety of basic and advanced circuits, including complex architectures, without any node or number of device limitations.

So even though SPICE has not gone over version 3, it has encouraged development through the Open Source forum. Companies that develop their



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of designs with efficient debugging. PSpice A/D also allows users to design and generate simulation models for transformers and DC inductors. Scalability options include PSpice Advanced Analysis capabilities and integration with MathWorks MATLAB Simulink for co-simulation," explains Madhavi Rao, senior marketing manager, Cadence.

Advanced capabilities such as temperature and stress analysis, electro-mechanical simulation, worst-case analysis, Monte Carlo analysis, and curve-fit optimisers are also available. This helps engineers design high-performance circuits that are reliable and can withstand parameter variation.

TINA-TI

TINA is a Spice-based circuit simulator

proprietary version of SPICE have ensured that it never really dies off.

To sum up

SPICE simulation has been used for over thirty years to accurately predict the behaviour of electronic circuits. Over the years, additional simulation algorithms, component models, bug fixes and other capabilities were added to the program. Besides these improvements, additions to the language have allowed simulation and modelling of more complex integrated circuits including MOSFETs. Moreover, numerous commercial versions of SPICE are now available. Even today SPICE is one of the most widely used circuit simulators in the world. ●

The author is a tech correspondent at EY Bengaluru

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